

Sample Size for Estimating Difference Between Two Means

1. Background

In an effort to effectively improve student academic achievement, the school plans to launch a new initiative (Initiative B). This program requires or encourages students to actively participate in specific extracurricular activities. The theoretical basis for this policy is that extracurricular activities foster discipline and teamwork, which in turn are expected to positively impact student test scores.

Before implementing this policy school-wide, teachers conducted a controlled experiment comparing two groups of students:

- **Group A (Control):** Students who focused solely on classroom lessons and did not participate in extracurricular activities.
- **Group B (Treatment):** Students who actively participated in extracurricular activities.

Assumptions & Known Parameters:

- Based on historical test score data from one month ago, the standard deviation (σ) of the test scores is 15 points.
- The minimum mean score difference considered practically meaningful (effect size / δ) is determined to be 5 points.
- The currently available real-world dataset includes a total of 4,234 students (2,287 control group students and 1,947 treatment group students).

The primary objective of this analysis is to determine the minimum sample size required to make the results of this experiment statistically reliable and to compare them with the currently available data.

2. Method to estimate the sample size

There are two approach to estimate the sample size:

Based on desired standard error

This approach focuses on limiting the level of uncertainty or precision of the interval estimate of the difference between two group means.

$$n = \left(\frac{2\sigma}{SE(\bar{y}_1 - \bar{y}_2)} \right)^2$$

- n is sample size estimate
- σ is standard deviation of population
- $SE(\bar{y}_1 - \bar{y}_2)$ is desired standard error of difference of two means

Based on Power and Confidence Level

This approach focuses on ensuring the ability of statistical tests to detect significant effects or differences in the field.

$$n = \frac{(2\sigma(z_{1-\frac{\alpha}{2}} + z_{1-\beta}))^2}{\delta^2}$$

- n is sample size estimate
- σ is standard deviation of population
- Z1 - α/2 represents z-value under desired significance level (α)
- Z1 - β represents z-value under the desired power
- δ The minimum difference you want to detect from the true population mean (effect size).

3. Sample Requirement Calculation Analysis Results

Scenario 1: Calculation Results Based on Standard Error (SE)

In the first scenario, a strict precision target is set where the standard error of the difference in scores must not exceed 2 points, assuming an initial σ estimate of 13 points.

$$n = \left(\frac{2 \times 13}{2}\right)^2$$

$$n = 169$$

- Total Sample Required: 169 students
- Sample Per Group (Balanced): 84.50 → 85 students

Scenario 2: Calculation Results Based on Hypothesis Testing (Power and Confidence)

The main experimental scenario established industry-standard research criteria: a 95% Confidence Level (type I error α = 5%) and 80% Statistical Power (80% power to detect a significant effect), with a σ of 15 points to detect a 5-point difference in effect (δ).

$$n = \frac{(2 \times 15 \times (1.96 + 0.84))^2}{5^2}$$

$$n \approx 283$$

- Total Sample Required: 283 students
- Sample Per Group (Balanced): 141.28 → 142 students

4. Sensitivity Analysis

A sensitivity analysis was conducted to understand how changes in the significance threshold (α) and test power (β) affected fluctuations in the minimum sample size requirement:

- Scenario A ($\alpha = 0.10$, Power = 80%): Lowering the confidence level to 90% reduces the minimum sample size to a total of 223 students (112 per group).
- Scenario B ($\alpha = 0.50$, Power = 95%): Extremely lowering the significance accuracy but increasing the detection target (Power) to 95% results in a total requirement of 194 students (97 per group).

5. Executive Summary & Managerial Recommendations

Measurement Approach	Parameter Criteria	Total Sample	Per Group	Information
Based on Standard Error	$\sigma = 13$, SE = 2	~ 170	~ 85	Focus on stability estimation
Based on Hypothesis Testing	$\sigma = 15$, $\delta = 5$, Conf = 95%, Power = 80%	~ 283	~ 142	Standard inference experiment
Sensitivity Scenario A	Conf = 90%, Power = 80%	~ 223	~ 112	More relaxed significance criteria
Sensitivity Scenario B	Conf = 50%, Power = 95%	~ 194	~ 97	Complete dominance in minimizing Type II Error

Strategic Recommendations for Decision-Making:

The minimum sample size required by statistics ranges from 142 to 300 students to provide valid conclusions. The fact that the school has 4,234 active students provides three significant strategic advantages:

- Near-100% Power: This experiment is guaranteed to detect even the smallest differences caused by extracurricular activities with great sensitivity.
- Very Low Standard Error: The estimated interval for the difference in mean scores will be very narrow and accurate.
- Subgroup Analysis Validity: The school can validly perform further analysis (e.g., analysis by gender, grade level, or type of extracurricular activity) without worrying about insufficient sample size.

Thus, the current database is very robust and more than sufficient to support accurate school policy decisions.